A toy hand, then a real one

How a Hand Works

Name _____ Date _____



Little Wonderer sets a mechanical hand toy on the table. Its fingers stand open, and five thin strings hang from the wrist. Big Wonderer leans in beside the little one. Your own hand can curl and open without any strings you can see. What is hiding inside it that works like the toy?



Each toy finger is a row of short, stiff pieces. A real finger is built the same way, with three small bones in a line. The thumb has two. Counting the wrist and palm, one hand holds 27 bones. Bones are hard, so they keep a finger's shape, but they cannot move on their own.



Pull one string, and one toy finger curls toward the palm. The string runs through small loops that keep it close to the finger. Your hand has strings too, called tendons. Muscles up in your forearm pull those tendons, and the tendons pull the finger bones. Tough loops along each finger hold the tendons close, just like the toy's loops.

bone	a hard, stiff part inside the body that holds its shape
tendon	a tough cord that joins a muscle to a bone
muscle	a body part that pulls to make bones move
joint	a place where two bones meet and can move
forearm	the part of the arm between the elbow and the wrist

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The toy finger bends in three places, and so does yours. Those bend points are joints, where two bones meet. Most finger joints work like a hinge on a door, folding mostly one way. The thumb's base joint is different. It lets the thumb swing across the palm to touch each fingertip.



They pull each string, one at a time, and draw a line from string to finger. The little one curls a real finger slowly and feels the forearm with the other hand. Something moves under the skin near the wrist. The toy and the hand follow the same plan: a pull, a string, and a joint that bends.



Let go of a toy string, and something must open the finger again. A muscle can only pull, never push. So your hand has a second set of tendons along the back of each finger, pulling it straight. Toys may use an elastic cord or a second string. Why might one of your fingers be hard to lift by itself?

1. What keeps each finger stiff?
2. What pulls a finger closed, in a toy and in a real hand?
3. Where does a finger bend, and how does the thumb move differently?
4. What opens a hand again, since muscles can only pull?

Your turn

Write or dictate three sentences: what keeps a finger stiff, what pulls it closed, and where it bends.

Example: Bones keep my finger stiff. A tendon pulls it closed. It bends at three joints.